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TECH HUB

• CRITICAL THINKING SERIES • NO. 03

# How to Spot Misinformation and Fake News

Your Practical Field Guide to the Information War.

No. 03 of 30 | Critical Thinking Series

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- Misinformation vs disinformation vs mal-information — defined
- The 7 types of information disorder (First Draft framework)
- The SIFT method — Stop, Investigate, Find, Trace
- How AI creates deepfakes and synthetic fake news at scale
- A 10-point checklist to verify any story before you share

# We Are Living Inside an Information War

Every day you encounter hundreds of pieces of information. Some are true. Some are false. Many are somewhere in between. The ability to tell the difference is no longer just a nice skill to have — **it is a survival skill** in a world where false information travels six times faster than true information (MIT Media Lab, 2018) and AI can now generate convincing fake news, images, and videos at industrial scale.

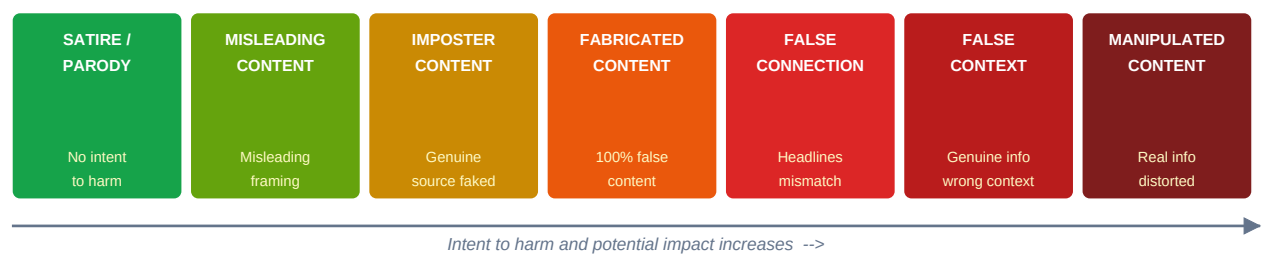
## Three Terms You Must Know

| Term            | Definition                                                       | Intent                                                                  | Example                                                                 |
|-----------------|------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Misinformation  | False or inaccurate information, regardless of intent to deceive | No deliberate intent — error, misunderstanding, satire shared seriously | A well-meaning person shares a statistic that turned out to be wrong    |
| Disinformation  | Deliberately false information created and spread to deceive     | Intentional — to manipulate opinion, undermine trust, cause harm        | A government creates fake news to discredit a political opponent        |
| Mal-information | True information used deliberately to cause harm                 | Intentional — uses real facts taken out of context                      | Publishing true but private information to destroy someone's reputation |

## The 7 Types of Information Disorder

Not all false information is the same. The First Draft framework (Claire Wardle, 2017) identifies seven distinct types, ranging from harmless satire to deliberately weaponised disinformation. The severity scale below shows how intent to harm increases across the spectrum.

Figure 1 — The 7 Types of Information Disorder (First Draft Framework)



| Type               | What It Is                                                                       | Real-World Example                                                             |
|--------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Satire / Parody    | Exaggerated humour with no intent to deceive — but can be mistaken for real news | The Onion headline shared by someone who did not realise it was satire         |
| Misleading Content | Genuine information used in a misleading way — cherry-picking facts              | A graph with a manipulated y-axis to make a small change look dramatic         |
| Imposter Content   | Genuine sources impersonated — fake BBC, CNN, or government websites             | A fake WHO website that looks identical to the real one, spreading health lies |
| Fabricated Content | Entirely false content — invented facts, fake quotes, made-up statistics         | A tweet falsely attributed to a public figure, screenshotted and shared        |
| False Connection   | Real headlines, images or captions that do not match the actual story            | A photo of a flood in one country used as "evidence" of a disaster in another  |
| False Context      | Genuine content shared with false contextual information                         | A real video of a protest from 2015 reshared as if it happened yesterday       |

|                     |                                                    |                                                                             |
|---------------------|----------------------------------------------------|-----------------------------------------------------------------------------|
| Manipulated Content | Real images or videos digitally altered to deceive | A deepfake video of a politician appearing to say something they never said |
|---------------------|----------------------------------------------------|-----------------------------------------------------------------------------|

# The SIFT Method — Your Four-Step Verification Tool

Developed by digital literacy expert Mike Caulfield, SIFT is the most practical, field-tested framework for quickly evaluating whether a piece of information is trustworthy. It takes 60–90 seconds and can be applied to any article, social media post, image, or video.

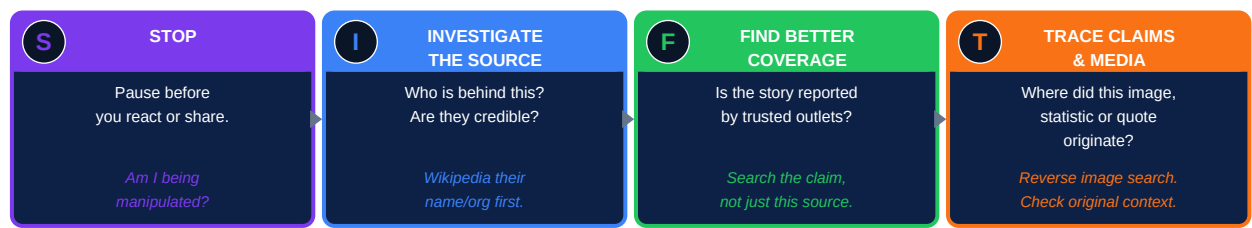


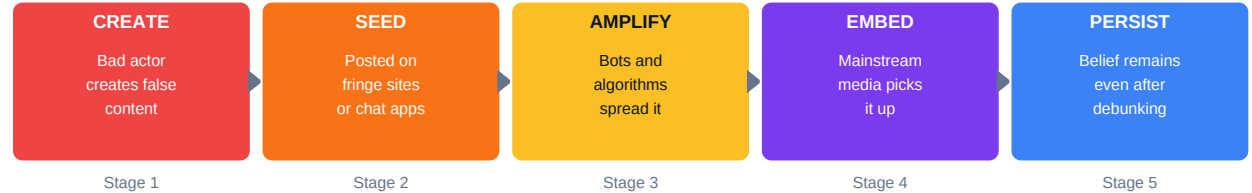
Figure 2 — The SIFT Method for Evaluating Information (Mike Caulfield, 2019)

**Reading laterally vs vertically:** Most people read an article deeply (vertically) before deciding if it is trustworthy. Expert fact-checkers do the opposite — they immediately open new tabs and check what OTHER sources say about the outlet or claim (reading laterally). This is consistently faster and more accurate.

## The Misinformation Lifecycle

False information does not spread randomly — it follows a predictable lifecycle. Understanding each stage helps you identify where a piece of misinformation is in its journey and how deeply it has already embedded itself into public belief.

Figure 3 — The Misinformation Lifecycle: From Creation to Persistent Belief



| SIFT Step                  | Tools to Use                                                                                               | Time   |
|----------------------------|------------------------------------------------------------------------------------------------------------|--------|
| S — Stop                   | Put down the urge to share. Ask: why does this make me feel strongly?                                      | 10 sec |
| I — Investigate the Source | Wikipedia the outlet/author. Check Media Bias/Fact Check (mediabiasfactcheck.com)                          | 60 sec |
| F — Find Better Coverage   | Google the core claim. Look for AP, Reuters, BBC, or specialist sources reporting the same story           | 60 sec |
| T — Trace the Original     | Reverse image: Google Images, TinEye. Trace quote to original context. Check archive.org for deleted pages | 90 sec |

**The "Backfire Effect" warning:** Research shows that correcting people's false beliefs sometimes strengthens those beliefs (Nyhan & Reifler, 2010). When you share fact-checks with others, focus on the facts — not on making someone feel foolish for believing a false story. Humiliation triggers defensiveness, not openness.

## Red Flags and AI-Generated Fake News

Before applying the full SIFT method, a quick scan for red flags can often identify misinformation in seconds. The following warning signs should trigger your critical thinking and slow your reaction.

| Red Flag              | What to Look For                                                                         | Why It Signals Risk                                                          |
|-----------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Emotional headline    | Words like SHOCKING, OUTRAGE, THEY WILL NEVER TELL YOU, caps lock, excessive punctuation | Designed to trigger emotional System 1 thinking and bypass critical analysis |
| No author byline      | Article with no named journalist, no credentials, no contact information                 | Credible outlets stand behind their work. Anonymity hides accountability.    |
| No publication date   | Undated articles, or a story being reshared without checking its original date           | Old stories are regularly recycled to make past events seem current          |
| URL and domain tricks | ABCnews.com.co   CNN-reports.net   BBC24news.org — slight misspellings of real brands    | Imposter sites deliberately mimic trusted brands to steal credibility        |
| Cite-less statistics  | "Studies show..." or "Experts agree..." with no named study or source                    | Real statistics come from named studies, institutions, or data sources       |
| Image mismatch        | Striking image that feels disproportionate to the story                                  | Real images are often stolen from other events, countries, or years          |
| Only one source       | Story reported by only one outlet, no corroboration from others                          | Major real events are covered by multiple independent news organisations     |

### The AI Misinformation Threat

| AI Tool                                       | What It Can Fake                                                                               | Detection Approach                                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Large Language Models (ChatGPT, Gemini, etc.) | Convincing fake news articles, fabricated quotes, invented statistics, fake research papers    | AI detectors (ZeroGPT, GPTZero) — imperfect but useful. Check all statistics independently.   |
| Deepfake Video (e.g. Sora, Kling)             | Realistic videos of politicians, celebrities, or ordinary people saying things they never said | Look for: unnatural blinking, lip-sync errors, background distortions. Check with InVID tool. |
| Synthetic Images (Midjourney, DALL-E)         | Fake news photographs — disasters, crowds, events that never happened                          | Reverse image search. Check for AI tells: hands, teeth, text in images are often wrong.       |
| Voice Cloning (ElevenLabs, etc.)              | Audio recordings faking someone's voice for phone scams, fake confessions                      | Verify through official channels. Call back on a known number. Check with audio tools.        |
| Automated Accounts (AI bots)                  | Fake social media accounts that amplify false content at scale to create false consensus       | Check account age, follower ratio, posting frequency. Real people have diverse histories.     |

### Your 10-Point Verification Checklist

| # | Check                                                     | Pass / Fail Signal                                                      |
|---|-----------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | Who published this? Is the outlet credible and real?      | PASS: Known outlet with real staff. FAIL: Unknown site, no staff page.  |
| 2 | Who wrote it? Does the author exist and have credentials? | PASS: Named journalist with a track record. FAIL: No byline, pseudonym. |

|    |                                                                     |                                                                                     |
|----|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 3  | When was it published? Is this story actually current?              | PASS: Recent, clearly dated. FAIL: Undated or old story reshared as new.            |
| 4  | What are the sources? Are statistics cited with named references?   | PASS: Linked to named studies/data. FAIL: "Studies show" with no citation.          |
| 5  | Is this covered by other credible outlets?                          | PASS: AP, Reuters, or BBC also report it. FAIL: Only this one source.               |
| 6  | Does the headline match the actual story?                           | PASS: Headline is accurate. FAIL: Misleading, exaggerated, or clickbait.            |
| 7  | Does the image actually match this story?                           | PASS: Reverse image search confirms context. FAIL: Image is from elsewhere.         |
| 8  | Is the URL real? Is the website design professional?                | PASS: Legitimate domain. FAIL: Misspelled brand name, poor design.                  |
| 9  | Am I sharing because it is true, or because it confirms my beliefs? | PASS: You verified it independently. FAIL: You are only sharing because you agree.  |
| 10 | Would I be comfortable if I was wrong about this?                   | PASS: You can admit error publicly. FAIL: Your identity is tied to this being true. |

What comes next — CT-004: The Socratic Method — Questioning Everything. How Socrates used dialogue to expose false certainty, and how you can use the same technique in conversations, self-reflection, and AI interactions today.